SM PHARMACEUTICALS SDN BHD

ANACAINE INJECTION
(LIDOCAINE HYDROCHLORIDE INJECTION)

DESCRIPTION:
A clear, colorless to pale yellow sterile solution

COMPOSITION:
For 1%: Contains Lidocaine Hydrochloride 10 mg/ml.
For 2%: Contains Lidocaine Hydrochloride 20 mg/ml.
Also contains Methyl Paraben 0.1% as preservative.

ACTIONS AND MODE OR MECHANISMS OF ACTION:
Block both the initiation and conduction of nerve impulses by reversibly stabilizing the neuronal membrane, thereby decreasing its permeability to sodium ions. This inhibits the depolarization phase of the neuronal membrane, resulting in the failure of a propagated action potential and consequent conduction blockade

PHARMACOLOGY (SUMMARY OF PHARMACODYNAMICS AND PHARMACOKINETICS):
Absorption: Complete systemic absorption.
Biotransformation: Xylidide metabolites are active and toxic.
Time to peak concentration: Usually 10-30 minutes.
Peak serum concentration: Depends upon factors influencing the rate of absorption from the injection site and rate of metabolism and distribution volume of the individual anaesthetic.
Elimination: Renal, primarily as metabolites. Quantity of dose excreted unchanged for Lidocaine is 10%.

INDICATIONS:
Local or regional anaesthetic solution for use in infiltration anaesthesia.

CONTRAINDICATIONS:
Hypovolaemia, complete heart block.
Known history of hypersensitivity to local anaesthetic of the amide type.
Do not use solutions containing adrenaline for anaesthetic in appendages.

SIDE EFFECTS / ADVERSE REACTIONS:
Hypotension, bradycardia, cardiac arrest, CNS effects include agitation, euphoria, respiratory depression, convulsion. Allergic reaction including cutaneous lesions, urticaria, oedema or anaphylactoid reactions. Systemic toxicity of local anaesthetic involve CNS and cardiovascular systems. There appear to be cross sensitivity between

ester and amide type local anaesthetics. Idiosyncrasy to local anaesthetics have been reported.
Hypersensitivity reactions to preservative (Methyl Paraben) in local anaesthetic preparations have been reported.

PRECAUTIONS / WARNINGS:
Risk – benefit should be considered when the following medical problem exist:
Congestive heart failure or Hepatic disease or impairment; Cardiovascular function impairment, especially heart block or shock; Drug sensitivity; History of or predisposition to malignant hyperthermia; Inflammation and/or infection in region of injection; Plasma cholinesterase deficiency – for esters; Renal disease; Myasthenia gravis; Epilepsy; and Local anaesthetic solutions containing preservative i.e. those supplied in MDV are not recommended for intrathecal anaesthesia.
Caution is also recommended in very young, elderly, acutely ill or debilitated patients, who may be more susceptible to systemic toxicity induced by local anaesthetics
Warning:
Lidocaine injection for infiltration and nerve block should be employed only be clinicians who are well versed in diagnosis and management of dose related toxicity and other acute emergencies that might arise from the block to be employed and then only after ensuring the immediate availability of oxygen. Other resuscitative drugs, cardio pulmonary equipment and the personal needed for proper management of toxic reactions and related emergencies.

Pregnancy / Reproduction
Local anaesthetic crosses the placenta by diffusion. The rate and degree of diffusion vary considerably amongst the various agents as determined by their rate of metabolism and physical characteristics such as plasma protein binding (reduced placental transfer with highly protein-bound agents), lipid solubility (greater placental transfer with nonionized form of agent).
First trimester – Retrospective studies of pregnant women receiving local anaesthetic for emergency surgery early in pregnancy have not shown that local anaesthetic cause birth defects. However, risk-benefit must be considered because the possibility of other foetal harm could not be excluded by these studies. Controlled studies in humans have not been done with any of these anaesthetics.
Breast-feeding
Although it is not known whether local anesthetics are distributed into breast milk, problems in humans have not been documented.

DRUG INTERACTIONS:
Beta adrenergic blocking agents
Concurrent use may slow metabolism of lidocaine because of decreased hepatic blood flow leading to increased risk of lidocaine toxicity.

Cimetidine
Cimetidine may inhibit hepatic metabolism of lidocaine leading to increased risk of lidocaine toxicity.
Antimyasthenics
(Inhibition of neuronal transmission by local anaesthetics may antagonize the effects of antimyasthenics on skeletal muscle, especially if large quantities of the anaesthetics are rapidly absorbed. Temporary dosage adjustment of antimyasthenics may be necessary to control symptoms of myasthenia gravis).
CNS depression – producing medication,
including those commonly used as pre-anaesthetic medication or for supplementation of local anaesthesia.
(Concurrent use with local anaesthetic may result in additive depressant effects. Caution and careful attention dosage of each agent are recommended).
Disinfectant solution containing heavy metals
(Local anesthetics may cause release of heavy metal ions from these solutions, which if injected along with the anesthetics, may cause several local irritation, swelling, and oedema. Such solutions are not recommended for chemical disinfectant of the container and preservative measures are recommended if they are used for skin or mucous membrane disinfection prior to anesthetic administration).
Monoamine oxidase (MAO) inhibitors including furazolidone, pargyline, and procarbazine
Concurrent use in-patients receiving local anesthetics via subarachnoid block may increase the risk of hypotension. Discontinuation of MAO inhibitors 10 days before elective surgery may be advisable if subaracnoid block anesthesia is planned.
Vasoconstrictors such as epinephrine, methoxamine or phenylephrine
(Use of methoxamine in combination with local anesthetics to prolong their action at local sites is not recommended, since methoxamine's extended effects may cause excessive restriction of circulation and lead to sloughing of tissue).
Other vasoconstrictors should be used cautiously and in carefully circumscribed quantities, if at all, with local anesthetics when anesthetizing are with end arteries (such as fingers, toes or penis) or with otherwise compromised blood supply, ischemia leading to gangrene may result).
Diuretics
(e.g Acetazolamide, loop diuretics and thiazides result in hypokalemia. This will antagonise the effects of lidocaine).

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:
The dosage is adjusted according to response of the patient and site of administration. The lowest and smallest dose producing the required effect should be given. The maximum dose for healthy adult should not exceed 200 mg.
Children and elderly or debilitated patients require smaller doses, commensurate with age and physical status.

Lidocaine Injection is used as a local anaesthetic when injected subcutaneously.

This solution is not intended for use intravenously. Solutions of lidocaine which contain preservatives should not be used for spinal, epidural, caudal or intravenous regional anaesthesia.
Route of administration are Subcutaneous (SC) or Intramuscular (IM)

SYMPTOMS AND TREATMENT FOR OVERDOSE AND ANTIDOTE(S):
Circulatory depression – Administering of a vasopressor (preferably ephedrine metaraminol, or mephentermine) and intravenous fluids is recommended. For maternal hypotension during obstetrical anesthesia, elevating the patient's leg or positioning the patient on her left side to displace the uterus may be sufficient to correct hypotension. Intravenous fluids and a vasoconstrictor should be administered if necessary; however, a vasoconstrictor should not be administered if the patients has received an ergot-type oxytocic agent.
Convulsions – If convulsions do not respond to respiratory support, Administering a benzodiazepine such as diazepam or an ultrashort acting barbiturate such as thiopental or thiamyla intravenously is recommended. The fact that these agents, especially the barbiturates, may cause circulatory depression when administered intravenously must be kept in mind. A neuromuscular blocking agent may also be used to decrease the muscular manifestations of persistent convulsions, artificial respiration is mandatory if such an agent is used.
Methaemoglobinaemia may be treated by the intravenous administration of a 1% solution of methylene blue.
Supportive care - Securing and maintaining a patent airway, administering oxygen, and instituting assisted or controlled respiration as required. In some patients, endotrachial intubation may be required.

Incompatibilities:
Lidocaine caused precipitation of Amphotericin, Methohexitone Sodium and Sulfadiazine Sodium in Glucose Injection. It is recommended that admixtures of Lidocaine & Glyceryl trinitrate should be avoided.

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:
Anacaine Injection 1%: Store below 30°C. Protect from light and freezing.
Anacaine Injection 2%: Store below 30°C. Protect from light and freezing.

PACKING / PACK SIZES:
Anacaine Injection 1%: 30 ml vial and 10 x 30 ml vials / carton
Anacaine Injection 2%: 10 ml vial and 10 x 10 ml vials / carton

SHELF LIFE:
3 Years

Manufacturer and Product Registration Holder

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